

MixED: A Mixed Entanglement Distribution Design for Efficient Quantum Teleportation in Quantum Communication Networks

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Abstract—Entanglement is the crucial resource for teleporting quantum bits (qubits) in quantum communication networks. However, the imperfection of entanglement swapping significantly degrades the rate and fidelity of long-distance entanglement distribution. To alleviate the negative effects of imperfect swap operations, this paper proposes a mixed entanglement distribution design, named MixED, aiming at improving the performance of qubit teleportation. Specifically, MixED mainly consists of two important schemes: the Entanglement Distribution Decision Scheme (EDDS) and the Entanglement Resource Allocation Scheme (ERAS). Based on entanglement distribution capability and the success probability of entanglement swapping, EDDS decides whether the memory-source-memory or memory-memory architecture is adopted to establish short-haul entanglement among three consecutive nodes under the guidance of a mixed strategy. This scheme guarantees the number of short-haul entanglements while effectively reducing the number of swap operations required on the selected path, thereby alleviating the degradation of rate and fidelity during long-distance entanglement distribution. To reduce the waste of entanglement resources caused by imperfect swap operations, ERAS adopts a priority-based strategy to determine the order in which each request is processed and then allocates high-fidelity entanglement resources in a greedy manner. ERAS improves resource utilization and thus enhances network throughput. Extensive simulations demonstrate that MixED outperforms four representative schemes regarding throughput, average fidelity, and the utilization of allocated entanglement resources, thus facilitating efficient qubit teleportation.

Index Terms—Quantum communication networks, entanglement distribution decision, entanglement resource allocation.

I. INTRODUCTION

AS THE promising platform, quantum communication networks follow the fundamental principles of quantum mechanics to achieve the *quantum bit (qubit)* transmission [1], [2], [3], thus supporting various ground-breaking quantum applications [4], [5]. As shown in Fig. 1, two important protocols, i.e., quantum teleportation and entanglement swapping, play a key role in qubit transmission. With the help of pre-shared entangled pairs, quantum teleportation allows for the information-theoretically secure transmission of data qubits over arbitrary distances [6], as shown in Fig. 1(a). Since almost all advanced or distributed quantum applications, such as distributed quantum computing and quantum-enhanced measurement, can have their core steps decomposed or modeled as a series of quantum teleportation processes, distributing entangled pairs between distant quantum nodes to support quantum teleportation becomes one of the important tasks in quantum communication networks [7]. However, it is challenging to directly establish entanglements between two distant nodes due to the inherent loss and noise in quantum channels. Quantum repeaters can be used to extend the range of entanglement distribution [8]. *Entanglement swapping* is the building block of quantum repeaters, enabling two never-interacted qubits to be entangled using Bell State Measurement (BSM) and classical communications [9], as shown in Fig. 1(b). In this way, two quantum nodes at any distance can establish entanglements by performing swap operations along a selected path consisting of multiple quantum repeaters.

However, swap operations are usually probabilistic due to the inherent limitations of quantum hardware [10], [11], [12]. Generally, the success probability of entanglement swapping refers to the probability that entanglement can be successfully established between two target nodes through BSM attempts at intermediate nodes [13]. The probabilistic nature of entanglement swapping significantly impairs long-distance entanglement distribution. *On the one hand, imperfect swap operations contribute to a rapid degradation in both rate and fidelity.* The rate of long-distance entanglement distribution is approximately proportional to the product of the success probability of entanglement swapping performed by quantum repeaters. Therefore, this rate decays rapidly as the

Received 4 March 2026; revised 20 May 2026; accepted 22 May 2026. Date of publication 1 June 2026; date of current version 10 June 2026. This work is supported in part by the Innovation Program for Quantum Science and Technology under Grant No. 2021ZD0301301, in part by the National Natural Science Foundation of China under Grant No. 62402466, the Natural Science Foundation of Anhui Province under Grant No. 2408085QF213, and Youth Innovation Promotion of the Association Chinese Academy of Sciences (CAS) under Grant No. Y202093. The associate editor coordinating the review of this article and approving it for publication was M. Caleffi. (Corresponding authors: Kaiping Xue; Hyundong Shin.)

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Digital Object Identifier 10.1109/TCOMM.2026.3698818

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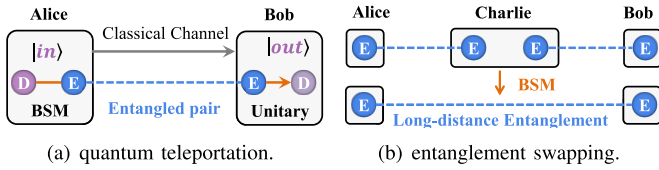


Fig. 1. The background of teleportation-based quantum communication.

number of swap operations increases, given a fixed number of entangled pairs available at each edge. The fidelity of the entangled pairs shared by distant quantum nodes also rapidly decreases with the number of swap operations [14], [15], and this degradation occurs because the fidelity of the entanglement established after each swap operation is approximately equal to the product of the fidelities of the two measured entangled pairs [16]. *On the other hand, imperfect swap operations cause the waste of entanglement resources.* Notably, entanglement resources cannot be reused due to the collapse-after-measurement phenomenon in quantum mechanics. In this case, failed entanglement swapping inevitably results in the waste of entanglement resources. As a result, the available entanglement resources decrease rapidly as the number of swap operations increases [17], thereby hindering entanglement distribution between multiple communicating parties. In summary, imperfect swap operations significantly hinder long-distance entanglement distribution.

To alleviate the negative effects of imperfect swap operations on long-distance entanglement distribution, some exploratory efforts have been made in the community. Existing studies mainly focus on routing optimization to select high-quality paths [18], [19], [20], [21], [22], [23], resource allocation optimization to improve network throughput [24], [25], [26], [27], [28], and studying the order of entanglement swapping to improve entanglement distribution rate [10], [12], [13], [29], [30], [31]. Although existing studies can select the path with lower hop counts and more available entanglement resources on each edge and improve the utilization of entanglement resources, they usually ignore the essential following fact about long-distance entanglement distribution. In quantum communication networks, there are usually three different entanglement distribution architectures in quantum communication networks [32], [33], i.e., Memory-Intermediate-Memory (MIM), Memory-Memory (MM), and Memory-Source-Memory (MSM). The architecture chosen by each quantum repeater to establish short-range entanglement affects the number of swap operations required on a repeater chain. Consequently, existing solutions often fail to achieve the expectation of high-performance entanglement distribution between distant nodes since the entangled pairs shared by quantum nodes cannot be efficiently utilized after multiple swap operations are performed.

Based on the important fact described above, this paper introduces MixED, a mixed entanglement distribution design, to support high-performance qubit teleportation by reducing the number of swap operations and improving resource utilization. First, we propose the Entanglement Distribution Decision Scheme (EDDS) to tackle the problem of the rapid degradation of both rate and fidelity caused by imperfect swap

operations. By balancing entanglement fidelity and the number of entangled pairs that can be successfully distributed, EDDS decides whether the MSM and MM architecture is chosen to achieve short-haul entanglement distribution among three consecutive nodes, thus establishing short-haul entanglement with heterogeneous link lengths instead of only single-link entanglements. In this case, EDDS can reduce the number of swap operations that need to be performed on a selected path while guaranteeing the success probability of entanglement swapping, thereby enhancing the performance of long-distance entanglement distribution in terms of rate and fidelity. Secondly, we introduce the Entanglement Resource Allocation Scheme (ERAS) to improve the utilization of entanglement resources and thus to enhance network throughput. ERAS realizes the allocation of heterogeneous entanglements among multiple requests to alleviate the waste of entanglement resources, thus increasing the number of entangled pairs finally shared by two communicating parties. We evaluate the performance of MixED using realistic and random network topologies and compare it against four representative schemes. Extensive simulation results demonstrate the superiority of MixED in terms of the throughput, the average fidelity of all established long-distance entanglements, and the utilization of allocated entanglement resources. In summary, the main contributions of this paper are listed as follows:

- We propose MixED, mainly consisting of EDDS and ERAS, to enhance long-distance entanglement distribution. EDDS adopts a mixed entanglement distribution strategy to reduce the number of swap operations, thereby alleviating the rapid degradation of rate and fidelity.
- After EDDS, ERAS models the allocation of heterogeneous entanglements among multiple requests as an optimization problem, and a priority-based request processing strategy and a greedy entanglement resource allocation algorithm are introduced to improve the number of entangled pairs finally shared by communicating parties.
- Extensive results show that MixED can enhance long-distance entanglement distribution in terms of throughput, average entanglement fidelity, and the utilization of allocated entanglement resources.

The rest of this paper is organized as follows. Section II introduces related work on long-distance entanglement distribution and presents our motivation. Section III presents the system model and problem statement. Section IV first presents the overview of MixED and then introduces two core schemes in detail. Section V performs extensive simulations to demonstrate the superiority of MixED. Finally, this paper is concluded in Section VI.

II. RELATED WORK AND MOTIVATION

A. Related Work

As the cornerstone of qubit teleportation, some exploratory schemes have been proposed to support entanglement distribution between a pair of source and destination nodes (hereafter referred to as an SD pair). To study or alleviate the negative effects of imperfect swap operations on long-distance

entanglement distribution, these studies primarily focus on path selection, entanglement resource allocation, and the order of entanglement swapping.

For the path selection problem, Van Meter et al. [18] explored the application of the Dijkstra algorithm to select a single path for each request; Pirandola et al. [19] proposed the entanglement entropy-enabled routing strategy in a diamond topology to select multiple paths; Pant et al. [34] adopted the greedy solution to present a multi-path routing by exploiting the diversity of multiple paths; Shi and Qian [20] proposed a routing algorithm named Q-CAST in a random graph, working by adding the path one by one with the largest expected throughput; Li et al. [21] proposed an entanglement routing algorithm, which guarantees that the fidelity of the established long-distance entanglements can meet the requirements of requests. Sutcliffe and Beghelli [22] studied three protocols that leverage multipath routing to increase the distribution rate of long-distance entanglement by dynamically selecting a path using knowledge of the successfully distributed entanglement states; Bugalho et al. [35] presented a multi-objective shortest-path-based algorithm for generating multipartite entanglement between different nodes with noisy quantum repeaters and imperfect memories; Yang et al. [23] proposed a stream processing-based routing scheme to reduce the average request completion time and further extend it with entanglement purification to provide a fidelity guarantee for long-distance entanglement distribution.

For the entanglement resource allocation problem, Li et al. [24] emphasized the importance of fairness and presented three representative entanglement resource allocation schemes for concurrent long-distance entanglement distribution; Chen et al. [25] introduced an entanglement resource allocation strategy to allocate limited resources for multiple requests based on probabilistic analysis and quasi-static modeling, ensuring a high level of fairness and efficiency from a network perspective; Zhao and Qiao [26] proposed the redundant entanglement resource provisioning and selection algorithm to maximize the network throughput, providing flexibility in allocating successfully created entanglement resources based on the network flow theory; Li et al. [27] presented the entanglement resource transfer and allocation schemes to alleviate the negative impact of resource scarcity on the request service capability while ensuring the fairness of entanglement resource allocation; Huang et al. [28] introduced an advanced resource allocation algorithm by viewing entanglement resource allocation as the problem of arm selection in adversarial group neural bandits.

For the problem of entanglement swapping order, Shchukin and van Loock [10] formulated the problem of finding the optimal entanglement swapping scheme as a Markov Decision Process (MDP) and presented its solution for different repeaters' sizes; Ghaderibaneh et al. [12] developed a dynamic programming algorithm to select an optimal swapping tree for a single pair of nodes under given constraints and an efficient iterative algorithm to compute a set of swapping trees for the general setting; Chang and Xue [13] showed that different entanglement swapping orders for the same path can lead to different expected throughputs; Iñesta et al. [29] introduced a general MDP model for homogeneous repeater

chains with memory cutoffs and designed optimal policies for minimizing the expected end-to-end entanglement delivery time; Sundaram and Gupta [30] presented a greedy algorithm that iteratively determines the best entanglement swapping to perform at each stage based on the current network; Mai et al. [31] developed an efficient heuristic based on the greedy selection of nodes to sequentially perform swap operations according to their swapping scores, thereby maximizing the path throughput.

Although existing studies have attempted to enhance long-distance entanglement distribution, they do not perform as expected since the mixed entanglement distribution method is ignored. In fact, two different entanglement distribution architectures can be adopted to support the generation of two-link entanglement among three consecutive quantum nodes. However, different strategies affect the number of short-haul entanglements and swap operations, thereby causing differences in path throughput. Therefore, the performance of long-distance entanglement distribution can be improved by optimizing the entanglement distribution decision on the selected path. In this case, we propose MixED, and our motivation is explained as follows.

B. Our Motivation

Before illustrating our motivation, we present the background of our motivation. In quantum communication networks, there are three basic architectures for implementing entanglement distribution between two quantum nodes [32], [33]. According to the type of physical devices configured on the node, we can refer to the three architectures as MIM, MM, and MSM, respectively. For the MIM architecture, an external Bell State Analyzer (BSA) is positioned between two nodes. After each node emits photons separated from entangled pairs, qubits stored in quantum memories configured in both nodes are entangled until reception of the return classical message sent by the BSA [36], [37], [38]. The MM architecture closely resembles the MIM architecture, and the main difference is that the BSA is an internal component of one of the nodes [39]. For the MM architecture, the type-II spontaneous parametric down-conversion scheme [40], [41] and the single-atom excitation scheme [42], [43] can be applied to support single-link entanglement distribution. No matter which scheme is used, the physical process of single-link entanglement distribution can be abstracted as a quantum node preparing entangled pairs and separating an entangled qubit from each entangled pair as a flying qubit to be transmitted to the neighboring node via quantum channels. Moreover, the physical process usually involves matter and photonic qubits, especially the single-atom excitation scheme. Specifically, the matter qubit is usually used for information processing/storage within a quantum node, and the photonic qubit, known as flying qubit, is responsible for transmitting the entangled qubit. For the MSM architecture, an entangled source is located between two nodes, and two nodes are configured with quantum memories and BSA to establish entanglement [44].

With the rapid development of quantum information technology, quantum nodes will have the ability to prepare, store,

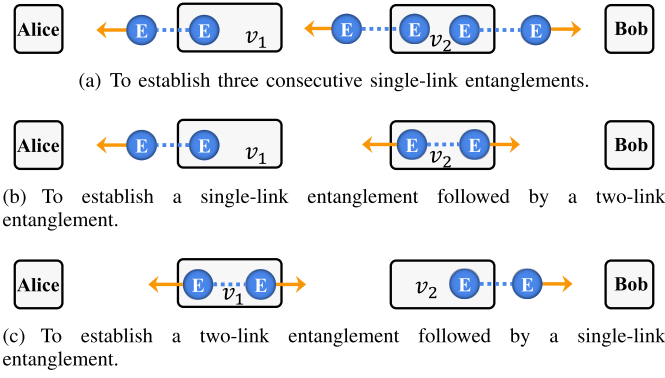


Fig. 2. An example of the mixed entanglement distribution design.

transmit, and manipulate qubits. In this case, both MM and MSM architectures can be used to achieve entanglement distribution among three consecutive nodes. We use Fig. 2 to illustrate our motivation for using the combination of MSM and MM architectures on a repeater chain. Alice and Bob are connected through two repeaters v_1 and v_2 , and there are two strategies to establish entanglement between Alice and Bob. For the first strategy, only the MM architecture is used, i.e., the repeater stores an entangled qubit in its local quantum memory and transmits another to one of its neighboring nodes. This strategy contributes to three consecutive single-link entanglements after short-haul entanglement distribution. For the second strategy, a mixed entanglement distribution method combining MM and MSM architectures is adopted. In this case, there are two types of short-haul entanglement on the path, that is, single-link entanglement established between two adjacent nodes and two-link entanglement established between two non-adjacent nodes spanning a single node. Combining these two strategies, we can derive three possible entanglement distribution decision results.

Fig. 2(a) shows the first decision, both v_1 and v_2 perform the functions of entanglement preparation and swapping to support the entanglement distribution between Alice and Bob. As a result, three consecutive single-link entanglements are established, and two swap operations need to be performed. When the channels' attenuation parameters are the same, switching the roles of v_1 and v_2 does not affect the quality of the three consecutive single-link entanglements since each pair of entangled qubits is distributed over the same distance. In the remaining two decision results, one repeater continues to perform the function it had in the first decision, while the other repeater only acts as the entangled source. The second decision is shown in Fig. 2(b), v_1 aims to establish single-link entanglement and v_2 distributes entangled pairs to v_1 and Bob to establish two-link entanglement. In this case, the entanglement distribution between Alice and Bob requires performing only one swap operation. The difference between the second and the last decisions is that the roles of v_1 and v_2 are switched, thereby establishing a two-link entanglement followed by a single-link entanglement, as shown in Fig. 2(c).

Since different entanglement distribution decisions contribute to differences in the capacity and fidelity of short-haul entanglement as well as the number of swap operations, these

three decisions vary in the performance of entanglement distribution between Alice and Bob. Consequently, it is possible to improve the performance of long-distance entanglement distribution by adjusting the role of each quantum repeater on the selected path. Inspired by this idea, we propose a mixed entanglement distribution design.

III. SYSTEM MODEL AND PROBLEM STATEMENT

A. Network Model

We formally model a quantum communication network as an undirected graph $G = (V, E)$, where V is the set of quantum nodes (including end nodes and repeaters) and E is the set of edges consisting of a quantum channel and a classical channel. Each quantum node is configured with a size-limited quantum memory for storing entangled qubits, thereby maintaining entanglement with other nodes. Referring to the development of the traditional Internet, this assumption aligns with the evolutionary logic of network architectures. In addition, each repeater can flexibly choose its own role on a selected path. The first type of role is a fully functional node that can realize entanglement preparation and entanglement swapping. The second role is simply to act as an entangled source, i.e., a repeater distributes two entangled qubits from an entangled pair to its predecessor and successor, respectively. In this case, two-link entanglement can be directly established without entanglement swapping.

B. Entanglement and Fidelity

A maximally entangled pair is a two-qubit entangled system in one of four Bell states (i.e., $|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$ and $|\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$) with four orthogonal basis states $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$. Bell states cannot be written as the product of two independent qubits. That is, a Bell pair is the maximally entangled state of two-qubit system since these two qubits are perfectly entangled. However, the interaction of Bell states with environmental noise destroys the perfection of maximally entangled systems. As a result, the Bell state will transform from a pure state into a mixed state after being distributed through noisy channels. In this work, we model an entangled pair in a mixed state as the Werner state [45].

Fidelity is an important metric used to measure the similarity between two quantum states. For any two entangled states ρ and σ , fidelity can be defined as $F(\rho, \sigma) = (\text{Tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}})^2$, where $\text{Tr}(\mathbf{A})$ is the trace of the parameter matrix $\mathbf{A}$. The inherent noise in quantum channels significantly affects entanglement fidelity. We characterize the channel noise existing in entanglement distribution as the depolarization model, as this is the worst-case assumption of channel noise. Let denote the noise probability as p , presenting the probability that an entangled state is disturbed or an error occurs. The effect of the depolarized channel on the two-qubit entangled state ρ can be expressed as

$$\sigma = \mathcal{E}(\rho) = (1 - p)\rho + p\frac{\mathbf{I}_4}{4}, \quad (1)$$

where σ is the Werner state after depolarization, and the noise probability p is essentially a function of the channel length L ,

i.e., $p(L) = 1 - e^{-\lambda L}$, where λ is the depolarization parameter that represents the depolarization intensity per unit length of the quantum channel. $\mathbf{I}$ refers to the two-dimensional identity matrix and $\mathbf{I}_4$ is equal to the inner product of $\mathbf{I}$ and $\mathbf{I}$. Here, we assume that the initial state of an entangled pair is a Bell state, i.e., $\rho = |\Phi^+\rangle$ and it is the pure state with a fidelity of 1.0. According to the definition of entanglement fidelity, we can derive the fidelity of the Werner state σ as

$$F(\sigma, |\Phi^+\rangle) = \langle \Phi^+ | \sigma | \Phi^+ \rangle = \frac{1 + 3e^{-\lambda L}}{4}. \quad (2)$$

Therefore, entangled pairs distributed via the same quantum channel are consistent in entanglement fidelity.

For the scenario where an intermediate repeater distributes an entangled pair to its predecessor and successor via two heterogeneous quantum channels, the state of the established entanglement is determined by the joint effect of the two channel noises. The noise probabilities of the two channels are $p_1 = p(L_1) = 1 - e^{-\lambda_1 L_1}$ and $p_2 = p(L_2) = 1 - e^{-\lambda_2 L_2}$, respectively. λ_1 and λ_2 are the depolarization parameters of the two channels, and L_1 and L_2 represent different channel lengths. In this case, suppose the initial state is $|\Phi^+\rangle$, the entangled state after suffering depolarization from two heterogeneous quantum channels can be expressed as

$$\rho_{12} = \gamma |\Phi^+\rangle \langle \Phi^+| + \frac{\delta}{4} \mathbf{I}_4, \quad (3)$$

where $\gamma = (1 - p_1)(1 - p_2)$ and $\delta = p_1(1 - p_2) + (1 - p_1)p_2 + p_1p_2$. According to the formula for fidelity, we can obtain the fidelity of ρ_{12} after decoherence. Specifically,

$$\begin{aligned} F(\rho_{12}, |\Phi^+\rangle) &= \langle \Phi^+ | \rho_{12} | \Phi^+ \rangle \\ &= (1 - p_1)(1 - p_2) + \frac{p_1(1 - p_2) + (1 - p_1)p_2 + p_1p_2}{4} \\ &= 1 - \frac{3}{4}(p_1 + p_2) + \frac{1}{2}p_1p_2. \end{aligned} \quad (4)$$

Consequently, the fidelity of the entanglement established by adopting the second strategy to distribute entangled pairs is jointly determined by two heterogeneous quantum channels.

C. Entanglement Distribution Model

Entanglement distribution between short-haul quantum nodes is a probabilistic process, and this probability generally decays exponentially with the channel length [46], [47]. In this work, we assume that quantum channels are identical except for their length. For the quantum repeater distributing entangled pairs to its predecessor and successor using the MSM architecture, the probability of successfully establishing a cross-hop entanglement is determined by the length of the longer quantum channel. Moreover, this probability is determined by the channel length in the case where the quantum repeater stores one entangled qubit and sends another to one of its neighbors using the MM architecture. After short-haul entanglement distribution, the number of entangled pairs shared by two quantum nodes is referred to as the entanglement capacity, which is an integer greater than 0.

Notably, the fidelity of an entangled state decreases over time due to couplings to the open environment, especially

noisy quantum memories. Generally, the decoherence process in noisy memories can be simulated using a white noise model in which a depolarizing operation is applied to the entangled state at every instant. In this case, the fidelity (denoted as $F(t)$) of a Werner state at time t can be formulated as follows:

$$F(t) = \frac{1}{4} + \left(F(t - \Delta t) - \frac{1}{4} \right) e^{-\Delta t/\tau}, \quad (5)$$

where Δt is an arbitrary interval of time and τ is a parameter that characterizes the exponential decay in fidelity of the whole entangled state, which is determined by the physical medium of quantum memories. According to [48], the typical lifetime of a qubit is 1.46 seconds, i.e., $\tau = 1.46\text{s}$.

In quantum communication networks, the success probability of entanglement swapping usually refers to the probability that entanglement can be successfully established between two target nodes through swap operations at intermediate nodes. The ability of a material system to distinguish Bell states is complete. That is, entanglement swapping is approximately deterministic. However, this deterministic property is the result of the assumption that “the initial two entangled pairs exist perfectly and the BSM is an ideal operation.” Therefore, the probabilistic nature of entanglement swapping is difficult to eliminate in practical quantum communication networks. We assume that the measurement fidelity of entanglement swapping on all repeaters is the same, denoted as η_{BSM} . In this case, the success probability of entanglement swapping is mainly determined by the fidelities of the two measured entanglements and η_{BSM} . Although this probability cannot be directly calculated by coupling these two factors, we can obtain it using mathematical statistics in real-world quantum networks, specifically by averaging the results of multiple entanglement swap attempts. In addition, the fidelity of the entanglement established after entanglement swapping is closely related to the fidelities of two swapped entanglements. As defined in [14] and [49], the fidelity of the new entanglement established after a swap operation can be approximately expressed by:

$$F_{new} = \frac{1}{4} \cdot \left(1 + (4\eta_{BSM}^2 - 1) \frac{(4F_1 - 1)(4F_2 - 1)}{3} \right), \quad (6)$$

where F_1 and F_2 denote the fidelity of the two entangled systems that were measured during entanglement swapping.

D. Problem Statement

To alleviate the degradation of rate and fidelity and the waste of entanglement resources, MixED needs to solve two key problems in such a general scenario where multiple SD pairs request to establish entanglements for teleporting qubits.

The first problem focuses on *determining the entanglement distribution strategy adopted by each repeater on the selected path*. The adopted strategy directly affects three important factors, i.e., the entanglement capacity, the number of swap operations, and the success probability of entanglement swapping, thus affecting how many long-distance entanglements an SD pair can theoretically establish along the selected path. To enhance the expected number of long-distance entanglements, MixED needs to determine the entanglement distribution

strategy adopted by each repeater. Specifically, this problem involves deciding whether each quantum repeater is responsible for establishing single-link or two-link entanglements to improve the long-distance entanglement distribution capability of a selected path. Thus, the first problem can be summarized as an optimization problem of maximizing the expected number of entanglements established along the selected path. We model the first problem and introduce its solution in Section IV-B.

The second problem involves *allocating limited and heterogeneous entanglement resources among multiple requests*. Notably, the resource allocation problem faced in MixED differs from the corresponding issue in existing studies. The reason is that a selected path contains both single-link and two-link entanglements after entanglement distribution, rather than only single-link entanglements. In addition, the entanglements established with the same link span exhibit variations in fidelity. Consequently, MixED cannot directly adopt existing solutions to implement entanglement resource allocation. The allocation of single-link entanglements versus two-link entanglements affects the success probability of entanglement swapping, thus affecting the performance of long-distance entanglement distribution. The goal of entanglement resource allocation is to make the most of the established entanglements for long-distance entanglement distribution, thus improving the number of entangled pairs shared finally by SD pairs. Consequently, we can model the entanglement resource allocation as an optimization problem, with its solution detailed in Section IV-C.

IV. MIXED ENTANGLEMENT DISTRIBUTION DESIGN

A. Overview

Similar to some representative works [20], [24], [50], MixED also adopts a time slot-based design to overcome the stochastic nature of entanglement distribution. In each time slot, MixED mainly consists of the following steps. First, multiple SD pairs spontaneously initiate requests to establish long-distance entanglements for qubit teleportation. Secondly, the shortest path routing algorithm is executed to select paths. Notably, considering the imperfection of entanglement swapping, two selected paths with small differences in hop count can show significant differences in path throughput. Moreover, the proposed scheme can reduce the number of swap operations to further amplify the impact of hop count on long-distance entanglement distribution. That is, our work can also significantly improve the entanglement distribution performance even when using the shortest path routing strategy. Therefore, we adopt the shortest path routing algorithm to select a path in this work. Thirdly, MixED executes the EDDS algorithm to determine the roles of quantum repeaters. That is, EDDS determines whether the MSM or MM architecture is chosen to achieve short-haul entanglement distribution among three consecutive nodes. According to the decision results of EDDS, entangled pairs are then prepared and distributed to establish either single-link or two-link entanglements. Fourthly, MixED performs the ERAS algorithm to allocate limited entanglement resources to multiple requests

to enhance the utilization of entanglement resources. Finally, swap operations are performed simultaneously to establish long-distance entanglements for teleporting qubits.

As the two core algorithms of the MixED scheme, EDDS and ERAS address different problems encountered when implementing a mixed entanglement distribution strategy on the selected path. The EDDS algorithm determines the optimal decision of two entanglement distribution roles for each node on a selected repeater chain to generate short-haul entanglements, while the ERAS algorithm allocates heterogeneous entanglement resources generated after the EDDS algorithm. The two algorithms address the problems that arise directly from combining different entanglement distribution architectures, and they work together to improve the performance of long-distance entanglement distribution.

B. Entanglement Distribution Decision Scheme

Different from existing studies that establish only single-link entanglement during short-haul entanglement distribution, EDDS adopts the combination of MSM and MM architectures to establish heterogeneous entanglements (i.e., single-link and two-link entanglements). Notably, entanglements with different link lengths vary in fidelity and quantity. Therefore, the decision result of EDDS directly affects the expected number of long-distance entanglements established along the selected path. Although the goal of EDDS, like existing research, is to maximize this expected value, the trade-offs between various factors increase the complexity of EDDS. To solve this problem, the EDDS algorithm is proposed and described in detail below.

Assume that the path with n quantum repeaters is selected for long-distance entanglement distribution between an SD pair. Mathematically, the selected path can be denoted as a set of quantum nodes belonging to $\mathbf{V}$, i.e., $\{v_s, v_1, \dots, v_n, v_d\}$, where v_i ($1 \leq i \leq n$) is the i -th quantum repeater on the selected path, v_s is the source node, and v_d is the destination node. The first task of EDDS is to split the selected path into multiple single-link and two-link segments, which corresponds to establishing either single-link or two-link entanglements. Assume that the selected path is divided into m segments, and let t be a variable with only two values to record the role decision results of all quantum repeaters. Specifically, $t_i = 0$ means that the i -th repeater is not a segmentation point, and $t_i = 1$ means that v_i is a segmentation point. If both t_i and t_{i+1} equal 1, the single-link entanglement is established between v_i and v_{i+1} . Otherwise, the condition that $t_i = 0$ and $t_{i+1} = 1$ represents v_i shares the two-link entangled pair with v_{i+1} . In this case, the problem of how to split a selected path to maximize the expected number of long-distance entanglements established between v_s and v_d can be formulated as

$$\text{Maximize } N_{sd} = C_{min} \cdot \prod_{i=1}^n p_{swap}^i, \quad (7)$$

$$s.t. \ C_{min} \leq C_j \in \mathbf{Z}^+, \forall j \in [1, m], \quad (7a)$$

$$t_i \in \{0, 1\}, \forall i \in [1, n], \quad (7b)$$

$$0 < p_{swap}^i \leq 1, \forall i \in [1, n], \quad (7c)$$

$$|S_j| \in \{1, 2\}, \forall j \in [1, m], \quad (7d)$$

$$\sum_{j=1}^m |S_j| = n + 2, \forall j \in [1, m], \quad (7e)$$

where $C_{\min} = \min\{C_1, C_2, \dots, C_m\}$, and C_j is the entanglement capacity of the j -th segment. p_{swap}^i is the success probability of the swap operation performed at the i -th quantum repeater. If $t_i = 0$, $p_{\text{swap}}^i = 1$; otherwise, p_{swap}^i is determined by the efficiency of measurement operations. In addition, S_j represents the j -th segment and $|S_j|$ is the hop count of the j -th segment. The first constraint reflects that the minimum capacity of short-distance entanglement directly determines the expected number of entangled pairs distributed through the selected path. The second constraint states that each quantum repeater on the selected path is either a sole or a dual repeater. Constraint (7c) reflects the imperfection of entanglement swapping. The last two constraints limit the hop count of each segment and ensure that the total hop count of all segments is equal to the selected path's hop count.

Algorithm 1 EDDS Algorithm

Input: The selected path $\{v_s, v_1, \dots, v_n, v_d\}$;
Output: The role of each quantum repeater;

- 1 $T \leftarrow \emptyset$;
- 2 $C_{\min} \leftarrow +\infty$;
- 3 $p_{\text{swap}}^{\text{all}} \leftarrow 1$;
- 4 **for each** repeater $i \in [1, n]$ **do**
- 5 $N'_{sd} \leftarrow \min\{C_{\min}, C_{(i-1,i)}\} \cdot p_{\text{swap}}^{\text{all}} \cdot p_{\text{swap}}^{i-1}$;
- 6 $N''_{sd} \leftarrow \min\{C_{\min}, C_{(i-1,i+1)}\} \cdot p_{\text{swap}}^{\text{all}} \cdot p_{\text{swap}}^i$;
- 7 **if** $N'_{sd} > N''_{sd}$ **then**
- 8 $C_{\min} \leftarrow \min\{C_{\min}, C_{(i-1,i)}\}$;
- 9 $t_{i-1} \leftarrow 1$;
- 10 $p_{\text{swap}}^{\text{all}} \leftarrow p_{\text{swap}}^{\text{all}} \cdot p_{\text{swap}}^{i-1}$;
- 11 **else**
- 12 $C_{\min} \leftarrow \min\{C_{\min}, C_{(i-1,i+1)}\}$;
- 13 $t_i \leftarrow 0$;
- 14 $p_{\text{swap}}^{\text{all}} \leftarrow p_{\text{swap}}^{\text{all}} \cdot p_{\text{swap}}^i$;
- 15 **if** $t_{n-1} = 0$ **then**
- 16 $t_n \leftarrow 1$;
- 17 **return** the decision results $T = \{t_i\}, \forall i \in [1, n]$.

For the problem described by Eq. (7), the most straightforward solution is to iterate over all possible results and then select the solution with the largest N_{sd} as the final output. Although this solution can find the optimal decision result, the computational complexity significantly increases with the number of quantum repeaters on the selected path. To reduce the computational complexity of finding the optimal solution, we transform the problem of how to divide a selected path into multiple segments into the problem of how to combine multiple segments hop-by-hop into a long-distance path. That is, starting from the first repeater, we add one quantum repeater at a time to construct a single-link or two-link segment until the last repeater is added. In this case, the computational complexity of each step of the solution process can be reduced from $\mathcal{O}(n)$ to $\mathcal{O}(1)$. The method of iteratively searching all possible segment results actually corresponds to solving the Fibonacci number counting problem, and its

computational complexity is $\mathcal{O}(\phi^n)$, where $\phi \approx 1.618$ [51], [52]. Therefore, EDDS reduce the computational complexity from an exponential level to a linear level.

Here, we introduce the EDDS algorithm to solve the problem described by Eq. (7). The selected path is denoted as $\{v_s, v_1, \dots, v_n, v_d\}$ and the decision result is recorded in the set T . The product of the success probabilities of all swap operations is denoted as $p_{\text{swap}}^{\text{all}}$ and initialized to 1. For each repeater, EDDS determines its role by comparing the number of long-distance entanglements that can be established when it acts as the endpoint of a single-link segment versus a two-link segment, respectively. Specifically, lines 5-6 calculate N'_{sd} and N''_{sd} , which denote the number of long-distance entanglements that can be established when the repeater is added to form a single-link and a two-link segment, respectively. If $N'_{sd} > N''_{sd}$, the role of v_{i-1} is determined, i.e., v_{i-1} is added to construct a single-link segment (Lines 7-10). Otherwise, v_{i-2} and v_i form a two-link segment and t_{i-1} is equal to 0 (Lines 11-14). The above process is performed iteratively until the last quantum repeater is added. Notably, each added quantum repeater can only determine the role of its predecessor. Therefore, the role of the n -th repeater is determined based on the decision result of v_{n-1} (Lines 15-16). Finally, EDDS returns a set containing the role decision results of each quantum repeater.

Theorem 1: The EDDS algorithm gives an optimal solution to the problem described by Eq. (7).

Proof: The mathematical induction method is adopted to prove this theorem. When the path contains only the source node v_s and the first repeater v_1 , the EDDS algorithm is optimal since there are no other choices. Assume that the sub-path P_{sub}^k contains the first k nodes (including the source node and $(k-1)$ repeaters), the decision result obtained by the EDDS algorithm is optimal, i.e., it maximizes $N_{(s,k-1)}$ of the sub-path. In this case, we assume that the minimum entanglement capacity is C_{sub}^{k-1} . Let add the $(k+1)$ -th node v_k to the path, forming the path $P_{\text{sub}}^{k+1} = \{v_s, v_1, \dots, v_{k-1}, v_k\}$. We need to prove that choosing v_k to form either a single-link segment with the previous node v_{k-1} or a two-link segment with v_{k-1} and v_{k+1} maximizes $N_{(s,k)}$ of P_{sub}^{k+1} . If v_k and v_{k-1} form a single-link segment, $N'_{(s,k)} = \min\{C_{(k-1,k)}, C_{\text{sub}}^{k-1}\} \cdot p_{\text{swap}}^{k-1}$, where p_{swap}^{k-1} is the success probability of the swap operation performed by v_{k-1} to couple the long-distance entanglement on sub-path and the single-link entanglement. Otherwise, three nodes, v_{k-1} , v_k and v_{k+1} , form a two-link segment, and $N''_{(s,k)} = \min\{C_{(k-1,k+1)}, C_{\text{sub}}^{k-1}\} \cdot p_{\text{swap}}^{k-1}$, where p_{swap}^{k-1} is the success probability of the swap operation performed by v_{k-1} to couple the long-distance entanglement on sub-path and the two-link entanglement. The EDDS algorithm chooses the larger of $N'_{(s,k)}$ and $N''_{(s,k)}$ as $N_{(s,k)}$ of the sub-path P_{sub}^{k+1} . Suppose there exists another decision result that yields a larger $\tilde{N}_{(s,k)} > \max\{N'_{(s,k)}, N''_{(s,k)}\}$. This means that $\tilde{N}_{(s,k)}$ is greater than both $N'_{(s,k)}$ and $N''_{(s,k)}$. However, according to the inductive hypothesis, $N_{(s,k-1)}$ and $N_{(s,k)}$ are the optimal value for the sub-paths P_{sub}^k and P_{sub}^{k+1} , respectively. This implies that $\tilde{N}_{(s,k)}$ would either need to improve the decision result of P_{sub}^k or improve the decision result of P_{sub}^{k+1} to obtain

a larger value, which contradicts the inductive hypothesis since $N_{(s,k-1)}$ and $N_{(s,k)}$ are already optimal values for their respective sub-paths. Therefore, the EDDS algorithm satisfies the optimal requirement.

C. Entanglement Resource Allocation Scheme

After EDDS and short-haul entanglement distribution, heterogeneous entanglements with different link lengths are established. Unlike existing methods, entanglement resource allocation in this work needs to consider not only entanglement capacity but also the link length and fidelity of entanglement. In this case, the design of the ERAS algorithm significantly affects the number of entangled pairs shared by each SD pair.

To improve the number of qubits teleported from source to destination nodes, ERAS needs to fully utilize available entanglement resources to establish as many long-distance entanglements as possible. Let denote the set of multiple requests in a time slot as $\mathbf{R}$, P_i is the path selected for the i -th request r_i , and S_i^j denotes the j -th segment of the i -th path. Correspondingly, the entanglement capacity and fidelity of the j -th segment are denoted as C_i^j and F_i^j . Besides, the number of entanglements allocated to the i -th request is denoted as A_i , and the number of entangled pairs finally shared by the i -th SD pair is N_{sd}^i . The goal of ERAS is to maximize the total number of entangled pairs shared by SD pairs in a time slot, and the entanglement resource allocation problem can be formulated as

$$\text{Maximize } \sum_{r_i \in \mathbf{R}} N_{sd}^i, \quad (8)$$

$$\text{s.t. } C_i^j, A_i \in \mathbf{Z}^+, \forall r_i \in \mathbf{R}, \quad (8a)$$

$$0 < F_i^j \leq 1, \forall r_i \in \mathbf{R}, \quad (8b)$$

$$0 \leq A_i \leq \min\{C_i^j\}, \forall r_i \in \mathbf{R}, \quad (8c)$$

Although some exploratory schemes have been proposed to address the above problem, existing schemes cannot be adopted here because entanglement resources generated in MixED are heterogeneous in quantity, hop span, and fidelity. Therefore, ERAS needs to design a new solution to realize the allocation of both single-link and two-link entanglements among multiple requests.

The results of EDDS cannot be adopted directly in ERAS since there may be partial overlap in the paths selected for two SD pairs, which may undermine the superiority of the results produced by EDDS. Since ERAS needs to consider the three properties of entanglement resources, namely quantity, hop span, and fidelity, the allocation of entanglement resources among multiple requests is an NP-hard problem. To solve the complex problem of entanglement resource allocation, ERAS iteratively allocates entanglement resources to one request at a time according to the priority of the selected path. Specifically, we divide ERAS into two steps, i.e., calculating the priority of each request and allocating limited entanglement resources to multiple requests in order of priority.

1) *Priority Calculation*: Here, we utilize the capability of long-distance entanglement distribution to characterize the priority of the selected path P_i , which is denoted as $W(P_i)$

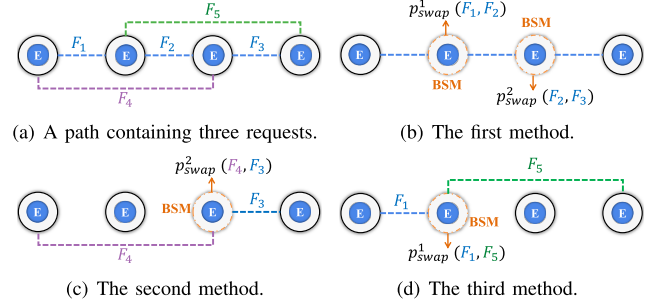


Fig. 3. An example of priority calculation. The first path consists of three single-link segments, the second path contains a single-link followed by a two-link segment, and the third path is the opposite of the second path. The fidelity of the single-link entanglement (blue color) are F_1 , F_2 , and F_3 , respectively. The fidelity of the two-link entanglement (red color) established in the second path is F_4 , while on the third path (green color) is F_5 .

and equals the product of the success probabilities of all swap operations that need to be performed. However, it is worth noting that the overlap between some paths selected for different requests leads to multiple possibilities for priority calculation on each path, and Fig. 3 is a typical example. As shown in Fig. 3(a), the paths selected for three different requests have partial overlaps on three consecutive edges, while the entanglement distribution decisions are distinct for each request. In this case, there are three different priority calculation results for the first request. The first is to use single-link entanglements on three single-link segments to extend the entanglement distribution distance, as shown in Fig. 3(b). In this case, the three-hop entanglement distribution capability of the path P_1 can be expressed as $W(P_1) = p_{swap}^1(F_1, F_2) \cdot p_{swap}^2(F_2, F_3)$. The second is to perform entanglement swapping at the first repeater by using two-link entanglement combined with the single-link entanglement on the last edge to establish three-hop entanglement, as shown in Fig. 3(c), and $W(P_1) = p_{swap}^2(F_4, F_3)$. Fig. 3(d) shows the third method, and $W(P_1)$ is equal to $p_{swap}^1(F_1, F_5)$.

In this work, the harmonic mean of all possible priority calculation results is adopted to evaluate the long-distance entanglement distribution capability of a selected path. The reason is that the harmonic mean is sensitive to smaller values, which is conducive to enhancing the performance of entanglement resource allocation in terms of fairness. In mathematics, the harmonic mean is defined as the reciprocal of the arithmetic mean of the reciprocals of the values. Therefore, the priority of the path P_i is expressed by

$$\tilde{W}(P_i) = \frac{m_i}{\sum_{j=1}^{j=m_i} \frac{1}{W_j(P_i)}}, \quad (9)$$

where m_i represents the number of all possible priority calculation results, and $W_j(P_i)$ is the j -th priority calculate result of the path P_i .

The single-link segments on each selected path can be divided into three categories. The first category is the independent single-link segments, which mainly refer to single-link segments that have no overlap with two-link segments or have overlap but cannot be converted to two-link entanglement by coupling with other single-link segments. The second category

Algorithm 2 Priority Calculation

Input: Three sets, i.e., paths $\mathcal{P}$, entanglement capacities $\mathcal{C}$, and segments $\mathcal{S}$;
Output: The result of priority calculation;

- 1 $\mathbf{W} \leftarrow$ Initialize the set of the path priority;
- 2 **for** each path $P_i \in \mathcal{P}$ **do**
- 3 $W^l(P_i) \leftarrow$ the capability of the left part of P_i ;
- 4 $F^l(P_i) \leftarrow$ the fidelity of the entanglement built via the left part of P_i ;
- 5 **for** $j = 1$ to m_i **do**
- 6 $W_j^r(P_i) \leftarrow$ the capability of the right part using the j -th method;
- 7 $F_j^r(P_i) \leftarrow$ the fidelity of the entanglement built via the right part using the j -th method;
- 8 $p_{sd} \leftarrow p_{\text{swap}}(F^l(P_i), F_j^r(P_i))$;
- 9 $W_j(P_i) \leftarrow W^l(P_i) \cdot W_j^r(P_i) \cdot p_{sd}$;
- 10 $j \leftarrow j + 1$;
- 11 $\tilde{W}(P_i) \leftarrow$ calculate the priority of the path P_i ;
- 12 Add $\tilde{W}(P_i)$ into the set of the path priority $\mathbf{W}$;
- 13 **return** the priority set $\mathbf{W} = \{\tilde{W}(P_i)\}, \forall P_i \in \mathcal{P}$.

is two consecutive single-link segments that overlap with a two-link segment, and the third category corresponds to the middle segment among three consecutive single-link segments and is the intersection of two two-link segments. In this way, the three types of single-link segments contribute to one, two, and three possible calculation results of long-distance entanglement distribution capacity, respectively. To simplify priority calculation, we can regard all independent single-link segments as the left part of the selected path and the rest of the single-link segments as the right part. Hence, we only need to focus on calculating all possible results of the path consisting of non-independent single-link segments.

Based on the above idea, we introduce **Algorithm 2** to calculate the priority of the selected paths, thereby guiding the decision on the order of resource allocation for each request. For the i -th selected path with m_i possible calculation results on the capability of long-distance entanglement distribution, this algorithm first calculates the capability of the left part of the path and the fidelity of the long-distance entanglement established by combining all independent single-link entanglements (Line 3-4). Then, the capability of the left part of the path is computed and combined with the right part to output the weight of the whole path (Lines 5-10). Specifically, for each combination method of the right part, the entanglement distribution capability of the path, as well as the fidelity of long-distance entanglement are first computed (Lines 6-7). When the capabilities of the left and right parts of the path are determined, the probability of connectivity between them directly affects the capability of the whole path. Therefore, the probability of connecting the two parts is first calculated, which is determined by the fidelities of the two long-distance entanglements established via the two parts of the path, and then the capability of the whole path is calculated (Lines 8-9). After all possible results are calculated, we can obtain the priority of the path P_i according to Eq. (9) (Line 11).

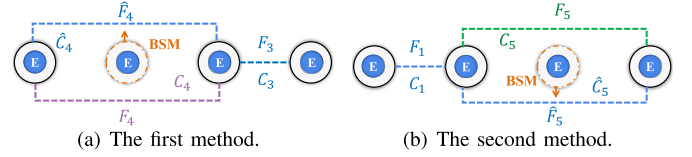


Fig. 4. An example of different resource allocation methods.

Finally, this algorithm outputs the priority set consisting of the priorities of all paths (Lines 12-13).

Here, we present the complexity analysis of **Algorithm 2**. Suppose there are n SD pairs requesting to establish long-distance entanglement, corresponding to n paths being selected. When the set of paths is determined, the time complexity of this algorithm is mainly decided by the number of methods to construct each path. Notably, the second category of single-link segment contributes two methods to construct each path, and the third category contributes three methods. Assuming that there are x two-link segments on each path that overlap with two consecutive single-link segments and y single-link segments of the third category, on average. In this case, there are m_i possible methods on average to construct the path $\mathcal{P}_i$, and $m_i = 2x + 3y$. Hence, the time complexity of calculating the weight of each path is $\mathcal{O}(x \cdot y)$. The time complexity of the whole algorithm in the time slot with n requests is $\mathcal{O}(x \cdot y \cdot n)$.

2) *Entanglement Resource Allocation*: According to the priority calculation results, we can allocate entanglement resources to each request in a descending order of priority. The resource allocation design needs to maximize the number of long-distance entanglements that can be established via each selected path. Notably, the existence of overlaps between different paths increases the complexity of resource allocation. Taking the scenario shown in Fig. 3(a) as an example, we assume that the entanglement capacity of each single-link segment is C_1, C_2 , and C_3 , and that the entanglement capacity of each two-link segment is C_4 and C_5 , respectively. Since two consecutive single-link entanglements can be converted to the two-link entanglement, there are two different methods to allocate entanglement resources on the path selected for the first request, as shown in Fig. 4. As a result, there are two types of two-link entanglements that differ in both quantity and fidelity: one is the original two-entanglement established by entanglement distribution, while the other is the converted two-link entanglement generated by combining two single-link entanglements via entanglement swapping.

As shown in Fig. 4(a), the first resource allocation method uses the single-link entanglements possessed by the first two single-segments to build the converted two-link entanglement. The number of the converted two-link entanglements is $\hat{C}_4$, and entanglement fidelity is $\hat{F}_4$. In this case, the number of three-hop entanglements that can be established using the first method can be expressed as

$$N_{three}^l = a \cdot C_4 \cdot p_{\text{swap}}^2(F_4, F_3) + b \cdot \hat{C}_4 \cdot p_{\text{swap}}^2(\hat{F}_4, F_3), \quad (10)$$

where $\hat{C}_4 = \min\{C_1, C_2\} \cdot p_{\text{swap}}^1(F_1, F_2)$. a and b are both decimals belonging to the set 0 to 1, which determine the number of original and converted two-link entanglements

Algorithm 3 Entanglement Resource Allocation

Input: Three sets, i.e., path $\mathcal{P}$, entanglement capacity $\mathcal{C}$, path priority $\mathcal{W}$;
Output: The results of resource allocation;

- 1 $\mathcal{W} \leftarrow$ Sort the priorities of the paths from high to low;
- 2 **for** each priority $\tilde{W}_i \in \mathcal{W}$ **do**
- 3 **for** each third-type single-link segment $S_i^k \in P_i$ **do**
- 4 $N_{three}^l \leftarrow$ combine the left neighbor;
- 5 $N_{three}^r \leftarrow$ combine the right neighbor;
- 6 **if** $N_{three}^l < N_{three}^r$ **then**
- 7 $\mathcal{C} \leftarrow \min\{C_i^k, C_i^{k+1}\} \cdot p_{swap}(F_k, F_{k+1});$
- 8 **else**
- 9 $\mathcal{C} \leftarrow \min\{C_i^{k-1}, C_i^k\} \cdot p_{swap}(F_{k-1}, F_k);$
- 10 $C_{min} \leftarrow$ the minimum entanglement capacity;
- 11 **for** each segment $S_i^j \in P_i$ **do**
- 12 **if** S_i^j is a two-link segment **then**
- 13 **if** $\hat{C}_i^j > C_{min}$ **then**
- 14 $A_i \leftarrow (C_{min}, \hat{F}_j);$
- 15 **else**
- 16 $A_i \leftarrow (C_{min}, \hat{F}_j, \check{F}_j);$
- 17 **else**
- 18 $A_i \leftarrow (C_{min}, F_j);$
- 19 Update the set of entanglement capacity;
- 20 **return** the allocation results $\mathcal{A} = \{A_i\}, \forall P_i \in \mathcal{P}.$

TABLE I
SIMULATION PARAMETERS

Parameters	Value
Global connection density	$\alpha = 0.6$
Distance attenuation factor	$\beta = 0.2$
Network area	$10^3 \text{ km} \times 10^3 \text{ km}$
Number of nodes	100-500
Number of SD pairs	5-25
Simulation rounds	200
Size of quantum memory	10-200 units
Depolarization parameter	$\lambda = 0.002$

allocated to the first request. The second method uses single-link entanglements possessed by the latter two single-link segments to build the converted two-link entanglement, as shown in Fig. 4(b). The number of three-hop entanglements that can be established using the second method is

$$N_{three}^r = a \cdot C_5 \cdot p_{swap}^1(F_5, F_1) + b \cdot \hat{C}_5 \cdot p_{swap}^1(\hat{F}_5, F_1), \quad (11)$$

where $\hat{C}_5$ is the number of the converted two-link entanglements and equals $\min\{C_2, C_3\} \cdot p_{swap}^2(F_2, F_3)$, $\hat{F}_5$ is the fidelity of the converted two-link entanglement. For the above method, if the two-link entanglement (e.g., $C_3 \geq C_4 + \hat{C}_4$) is the bottleneck resource, it needs to be fully utilized to extend entanglement distribution distance. In this case, both a and b are equal to 1. Otherwise, a and b are two variables that only need to satisfy resource constraint (e.g., $a \cdot C_4 + b \cdot \hat{C}_4 = C_3$).

By comparing Eq. (10) and Eq. (11), we can observe that two important factors affect long-distance entanglement

distribution. First, the resource allocation method determines which single-link entanglements are converted to two-link entanglements, thereby affecting the minimum entanglement capacity and the fidelity of the allocated entanglement. Secondly, the parameters a and b directly determine the utilization of the allocated two-link entanglement when single-link entanglement is the bottleneck resource, because the original and converted two-link entanglements are different in fidelity. The resource allocation design needs to first make full use of the two-link entanglement with higher fidelity. Moreover, the first factor, i.e., the resource allocation method, directly determines whether a and b are variables or fixed values, affecting how the original and converted two-link entanglements are used. To maximize the number of entangled pairs distributed via the selected path, the resource allocation design needs to first decide how to use the third type of single-link entanglements and then fully utilize the two-link entanglement with higher fidelity.

For the third type of single-link entanglement, we select the resource allocation method that can build more three-hop entanglements with higher fidelity to decide how they are allocated. Then, the entanglements possessed by two consecutive single-link segments are utilized to generate the converted two-link entanglement. After this, only the allocation of the original and converted two-link entanglements needs to be addressed. Here, we introduce the ERAS algorithm as shown in **Algorithm 3**. The algorithm first sorts the priorities and initializes the set of resource allocation results (Line 1). Then, the algorithm iteratively allocates entanglement resources along the path greedily to higher-priority requests and updates the set of entanglement capacities (Lines 2-19). For each selected path, two different resource allocation methods for the third-type single-link entanglement are compared, and the method that can establish more three-hop entanglements with higher fidelity is selected (Lines 3-9). Then, we use the minimum entanglement capacity (denoted as C_{min}) as the number of entanglements that the selected path allocates to the request (Lines 10-18). For each two-link segment, this algorithm preferentially utilizes the high-fidelity entanglement (Lines 12-16). If the number of two-link entanglements (denoted as $\hat{C}_i^j$) with a high fidelity (denoted as $\hat{F}_j$) is less than C_{min} , the two-link entanglement with a low fidelity ($\check{F}_j$) needs to be allocated to supplement the missing resources (Lines 15-16). After resource allocation on a path is completed, this algorithm updates the set of entanglement capacities (Line 19). Finally, Line 20 outputs the set $\mathcal{A}$ as the results of the entanglement resource allocation.

Here, we present the complexity analysis of **Algorithm 3**. We assume that there are n paths selected for long-distance entanglement distribution. After two consecutive single-link entanglements are converted into two-link entanglements, the average number of segments contained in each path is assumed to be y . For each selected path, the time complexity of resource allocation is mainly determined by the number of segments it contains. In this, the time complexity introduced by the resource allocation on each selected path is $\mathcal{O}(y)$. Therefore, the time complexity of the whole algorithm in the time slot with n requests is $\mathcal{O}(n \cdot y)$.

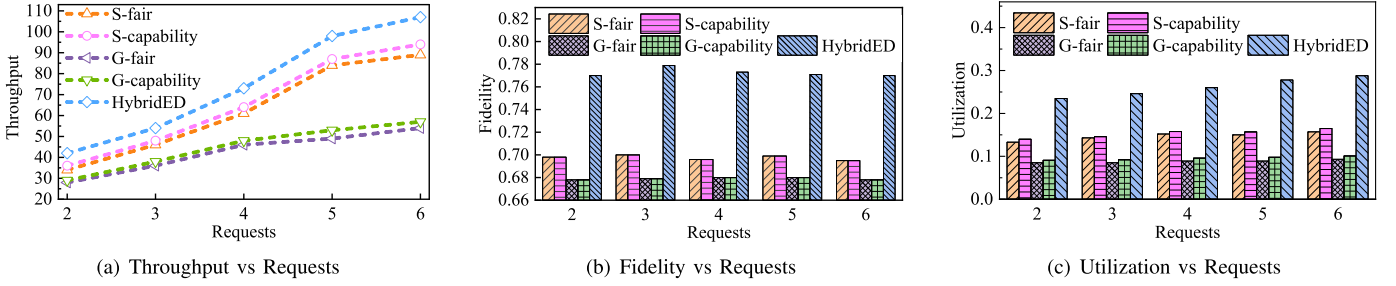


Fig. 5. The effects of requests' number on long-distance entanglement distribution using USNET network topology.

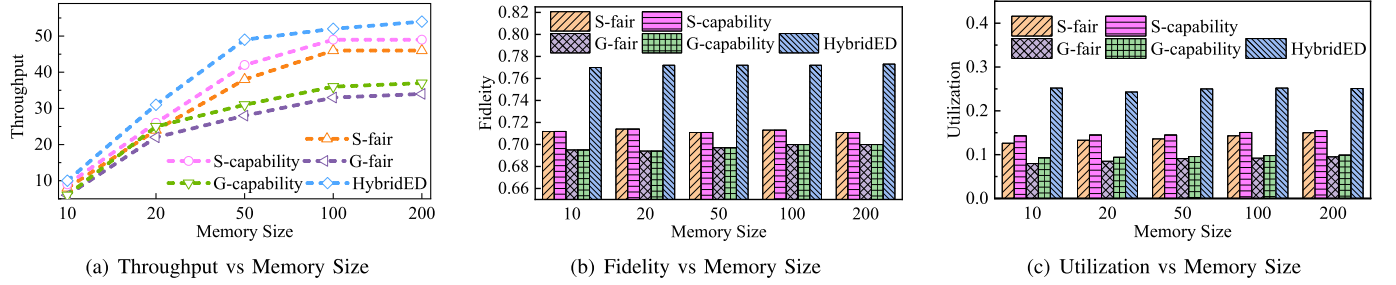


Fig. 6. The effects of memory size on long-distance entanglement distribution using USNET network topology.

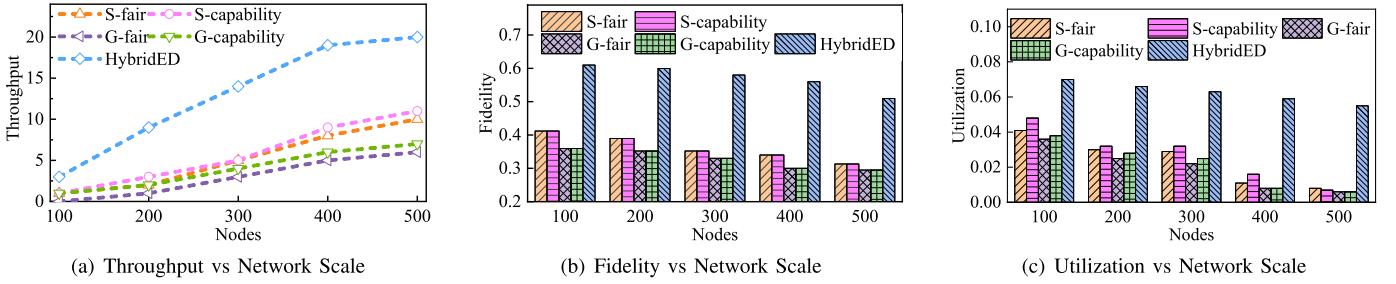


Fig. 7. The effects of network scale on long-distance entanglement distribution using a random network topology.

V. PERFORMANCE EVALUATION

A. Simulation Methodology

To evaluate MixED, we first use the USNET network topology that contains 24 nodes and 43 edges and then randomly place quantum nodes into a square area to generate a network topology. In the randomly generated topology, the Waxman model is adopted to determine the probability that two different quantum nodes are directly linked [53]. Simulation parameters are set as shown in Table I, and we show the average results of 200 simulation experiments.

We evaluate MixED using three metrics. One is the number of entangled pairs that can be distributed between SD pairs in a time slot, which is denoted as throughput. The second metric is the average fidelity of long-distance entanglements. A larger average fidelity is beneficial for qubit teleportation. The last metric is the utilization of the entanglement resources allocated on selected paths, and a larger utilization means that the allocated resources are less wasted. We adopt four schemes as comparison objects, and the core implementation of these comparison schemes has been widely adopted in related work, such as in [18], [20], [21], [24], [27], [54], and [55]. In terms of path selection, four comparison schemes

can be divided into two categories: one is the routing scheme based on the knowledge of the global topology, and the other is the shortest path routing scheme. The routing scheme that utilizes global topology knowledge is adopted to select the path with the highest expected throughput, which is similar to the well-known Q-PASS/Q-CAST routing algorithms introduced in [20]. In terms of resource allocation, the comparison schemes consist of two categories: one is the scheme for fairly allocating entanglement resources among multiple requests, while the other is the scheme that allocates entanglement resources based on the path capability. We denote these four comparison schemes as the S-fair, S-capability, G-fair, and G-capability, respectively.

- **S-fair:** The shortest path routing algorithm is adopted to select a path, and the fair-based scheme is used to allocate entanglement resources.
- **S-capability:** The shortest path routing algorithm is adopted to select a path, and the path-capability-based scheme is used to allocate entanglement resources.
- **G-fair:** The routing scheme based on the knowledge of the global topology is utilized to select a path, and the fair-based scheme is used to allocate entanglement resources.

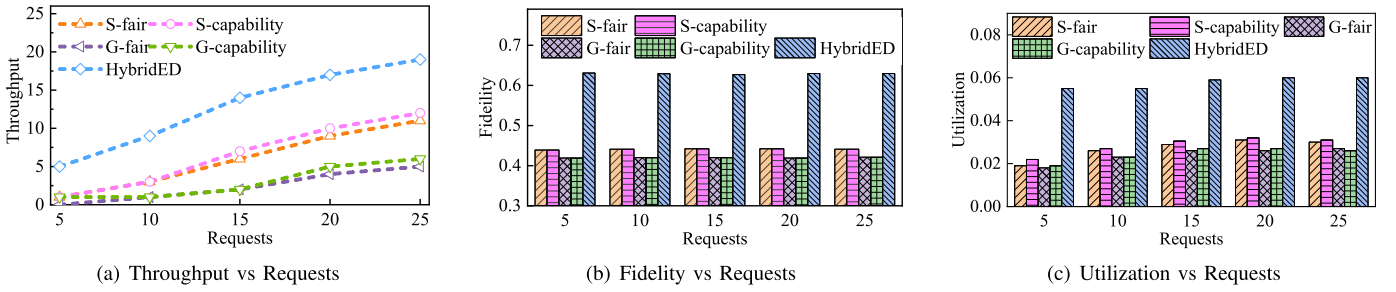


Fig. 8. The effects of requests' number on long-distance entanglement distribution using a random network topology.

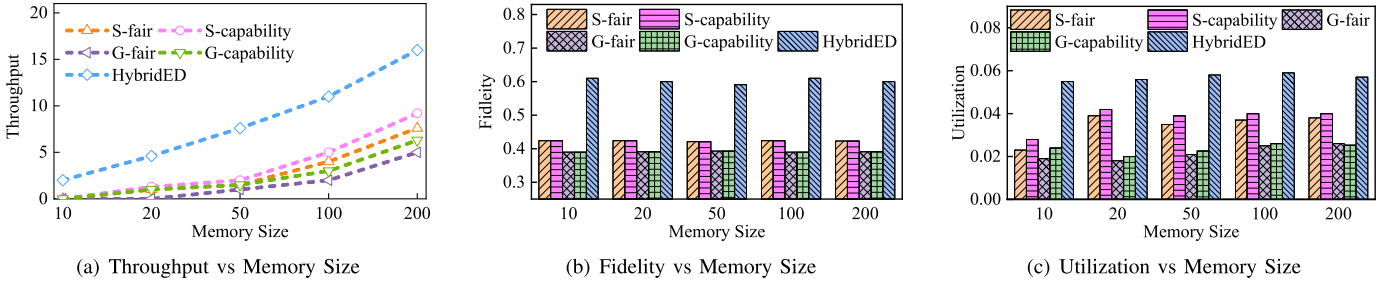


Fig. 9. The effects of memory size on long-distance entanglement distribution using a random network topology.

- **G-capability:** The routing scheme based on the knowledge of the global topology is utilized to select a path, and the path-capability-based scheme is used to allocate entanglement resources.

B. Evaluation in USNET

From a practical point of view, we first evaluate MixED using the USNET topology with the channel length reduced by a factor of 10. Simulation results demonstrate the superiority of MixED, as shown in Fig. 5 and Fig. 6.

Fig. 5(a) shows that five schemes can improve throughput as the number of requests increases. However, MixED performs the best, followed by the schemes that use the shortest path routing algorithm. The reason is that MixED can reduce the number of swap operations while balancing entanglement capacity and the success probability. As shown in Fig. 5(b), the comparison schemes that use the same path selection scheme exhibit consistency in the average fidelity because they are the same in the fidelity of the allocated entanglement on each edge and the number of swap operations. However, MixED considers the diversity of entanglement fidelity and can reduce the number of swap operations, thus effectively improving the average fidelity. As shown in Fig. 5(c), the comparison schemes that use the capability-based method show superiority over the fair-based method in resource utilization. This is because the capability-based method considers the probabilistic nature of entanglement swapping. MixED further reduces the number of swap operations, making it perform the best regarding resource utilization. Besides, more heterogeneous entanglement resources can be used to reduce the number of swap operations as the number of requests increases, and the resource utilization of MixED is slightly improved.

Fig. 6 shows the effect of memory size on five schemes when the number of requests is set to 4. As shown in Fig. 6(a),

the throughput of the five evaluated schemes first grows rapidly and then increases slowly as the memory size increases. When the memory size is limited (i.e., the memory size is smaller than the number of entangled pairs directly distributed between quantum nodes), it constrains the entanglement capacity of each edge, which in turn limits the throughput. However, regardless of the size of quantum memory, MixED is always the best-performing scheme since it can effectively mitigate the negative impact of probabilistic entanglement swapping. In terms of the average fidelity and the utilization of the allocated entanglement resources, the performance of the five evaluated schemes is generally consistent with the results observed in the simulation regarding the impact of requests' number, as shown in Fig. 6(b) and Fig. 6(c). This is because the size of quantum memory does not affect the fidelity of the allocated entanglement resources on each selected path and thus does not affect the success probability of entanglement swapping.

C. Evaluation in the Random Topology

From the perspective of scalability, we further vary network sizes, requests' number, and memory size to compare MixED with four representative schemes, using the random network topology. The simulation results are shown in Fig. 7, Fig. 8, and Fig. 9, respectively.

Fig. 7(a) shows the effect of network scale on throughput. The expansion of the network scale increases the hop count of each selected path, thereby amplifying the negative impact of probabilistic entanglement swapping. Therefore, there is a clear performance gap between the comparison schemes that use different path selection designs, and MixED performs best since it can reduce the number of swap operations. As shown in Fig. 7(b), the comparison schemes that use the same path selection design exhibit consistency in average fidelity. The average fidelity output by five schemes decreases with the

increase in network size since the increase in the hop count of each path contributes to the rapid decay of the average fidelity. However, MixED's performance decays slowly and always performs well. Besides, the increase in network size leads to an increase in the number of swap operations and the entanglement resources allocated along the path. In this context, more entanglement resources are allocated but more entanglement resources are wasted due to failed swap operations. Therefore, the performance of MixED is reduced in terms of resource utilization, as shown in Fig. 7(c). However, MixED is still superior to the other four comparison schemes.

Fig. 8(a) shows that the throughput increases as the number of requests increases, but the rate of increase gradually slows down. This is because the increase in the number of requests leads to an increase in the number of requests that share a single edge, which intensifies resource competition. Therefore, the number of entanglement resources that can be allocated to each request on a selected path is reduced, thereby slowing down the growth rate of throughput. In terms of the average fidelity, the performance of the five schemes is consistent with that of the USNET topology in terms of change trends. However, the average fidelity produced by each scheme is lower than that shown in Fig. 5(b) since the increase in network size results in an increase in the hop count of each selected path. As shown in Fig. 8(c), the other four schemes perform poorly in resource utilization because a larger path's hops cause more entanglement resources to be wasted. However, MixED still performs well since it can reduce the number of swap operations while ensuring the success probability of entanglement swapping.

As shown in Fig. 9(a), there is a significant difference between MixED and the four comparison schemes when the memory size of each edge is large enough. The reason is that more entanglement resources need to be allocated to requests on the paths with a larger number of hops in order to offset the negative impact of probabilistic entanglement swapping. Therefore, the quantum nodes with small memory size limit the improvement of long-distance entanglement distribution. However, MixED can reduce the number of swap operations while ensuring the success probability of entanglement swapping, so it promotes the improvement of throughput on the paths with limited memory size and large hop counts. As shown in Fig. 9(b), the results of the average fidelity are consistent with other simulation results since the memory size does not affect the fidelity of the entanglement resources allocated on each edge and the number of swap operations. In addition, MixED can improve the average fidelity by reducing the number of swap operations. In terms of resource utilization, the performances of the five schemes are basically consistent with the results exploring the impact of SD pairs' number on long-distance entanglement distribution, as shown in Fig. 9(c). This is because the increased memory size does not affect the number of swap operations and the success probability of entanglement swapping.

VI. CONCLUSION

In this paper, we proposed a mixed entanglement distribution design named MixED, which mainly consists of

two core schemes, i.e., EDDS and ERAS. EDDS alleviates the rapid degradation of both rate and fidelity by reducing the number of swap operations, using a mixed strategy to establish both single-link and two-link entanglements along the selected path. ERAS solves the problem of entanglement resource competition between concurrent requests to maximize network throughput by using priority calculation and entanglement resource allocation algorithms. Extensive simulations performed in real-world and random network topologies illustrate the advantages of MixED in terms of the rate and fidelity of long-distance entanglement distribution and in the utilization of entanglement resources, thus promoting efficient qubit teleportation in quantum communication networks.

ACKNOWLEDGMENT

Part of the work was completed in collaboration with Kyung Hee University, Republic of Korea.

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